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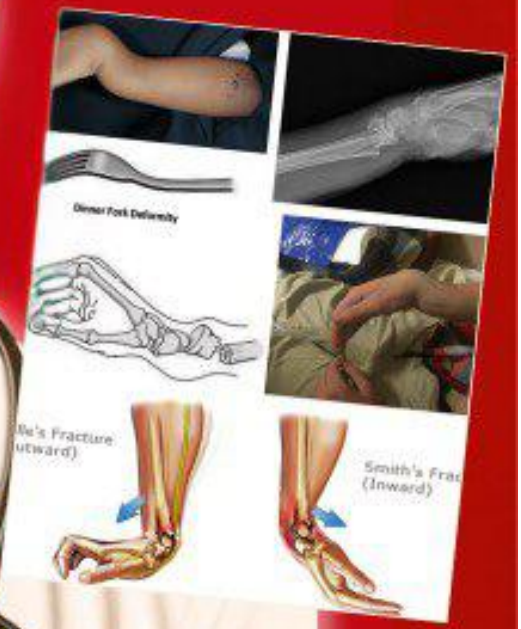
ANATOMY

REASONING SERIES

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CASE





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REASONING QUESTIONS ANATOMY
UPPER LIMB

1. What is the deformity seen in Winging of scapula? Explain with its anatomical basis.

Definition:

Winging of scapula is a condition where the **medial border of scapula** becomes **prominent** and moves **backward like a wing**, especially when the person pushes against a wall.

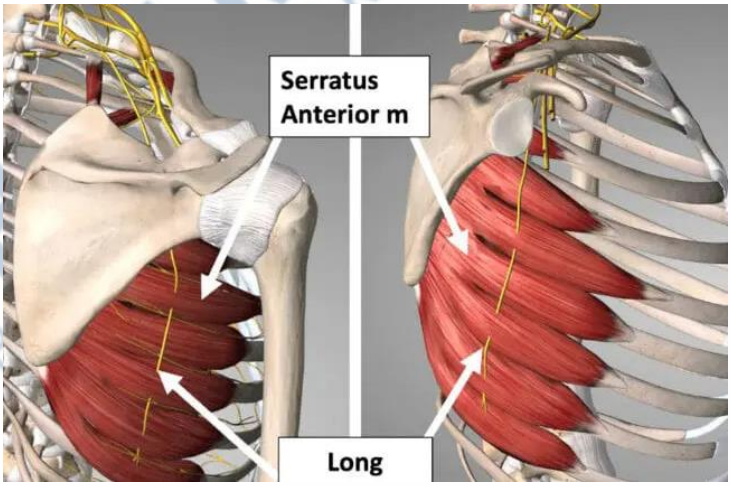
Cause:

Injury to the **long thoracic nerve** → paralysis of **serratus anterior muscle**.

Common reason:

During surgery, trauma, or carrying heavy loads on the shoulder.

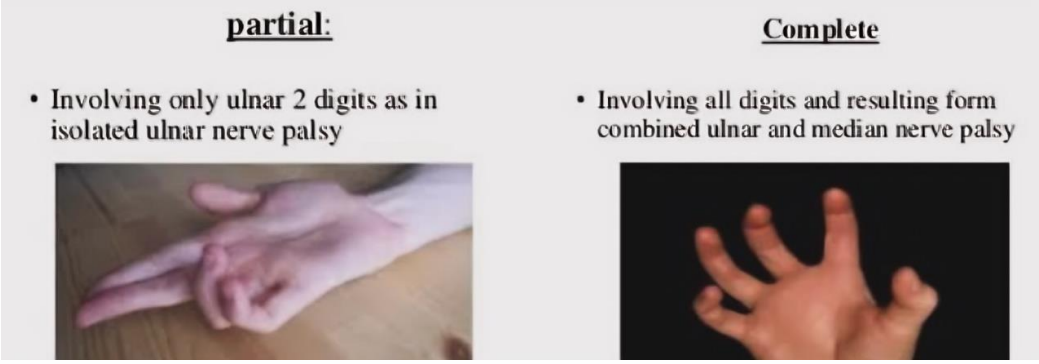




2. What is the deformity seen in Claw Hand? Explain with its anatomical basis.

Definition:

Claw hand is a deformity where the fingers are bent at the interphalangeal joints and hyperextended at the metacarpophalangeal joints.



Cause:

Injury to the **ulnar nerve**, especially at the wrist.

Affected Muscles:

Lumbricals and interossei muscles of the hand (mainly for ring and little finger).

Features:

- Hyperextension at MCP joints
- Flexion at IP joints
- Hollow spaces between metacarpals (due to muscle wasting)

Seen in: Ulnar nerve palsy, leprosy, trauma.

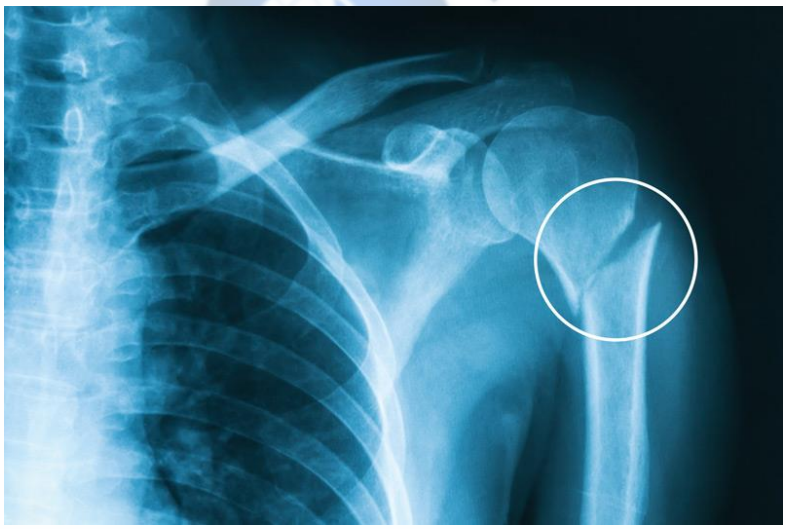


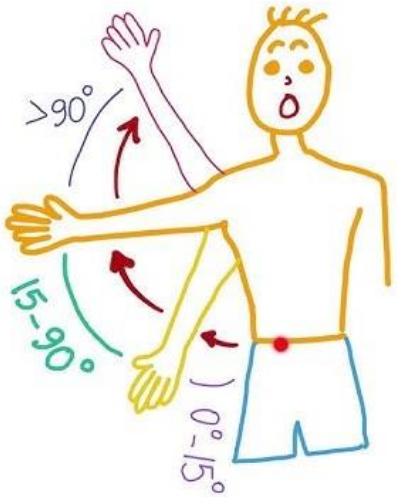
Claw hand of 2nd and 3rd digits
Median nerve

Claw hand of 4th and 5th digits ("Pope's blessing")
Distal ulnar nerve lesion

Klumpke's total claw hand
Lower trunk (C8, T1)

3. A 55-year-old woman fell on her outstretched hand and sustained a fracture at the surgical neck of her humerus. On examination, she had difficulty in abducting her arm beyond 15°, and there was loss of sensation over the lateral aspect of the shoulder





i) Nerve likely injured:

- **Axillary nerve**
- The **axillary nerve** winds around the **surgical neck of the humerus** along with the **posterior circumflex humeral artery**.
- Any fracture at this site endangers the axillary nerve, making it the most likely injured structure.

ii) Muscles supplied by that nerve:

- **Deltoid** and **Teres minor**
- **Deltoid muscle** – chief abductor of the arm from 15° to 90°.
- **Teres minor muscle** – helps in **lateral rotation** of the arm.
- Thus, axillary nerve injury compromises both abduction (deltoid) and lateral rotation (teres minor).

iii) Muscle affected:

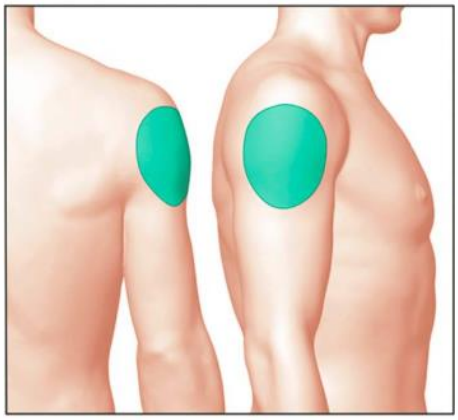
- **Deltoid** – leads to difficulty in abducting the arm (15° to 90°)
- The clinical sign was difficulty in abducting the arm beyond 15°

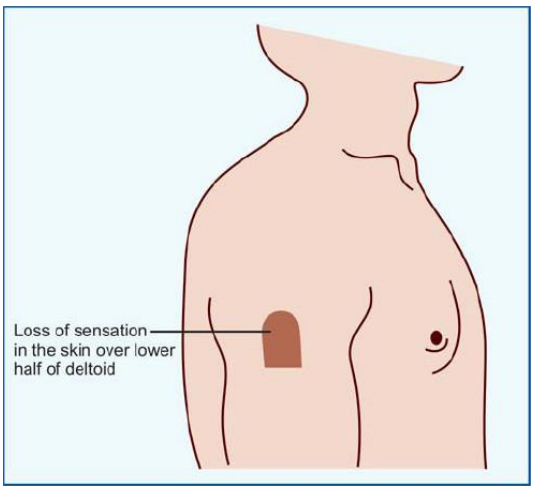
- First 0°–15° abduction is initiated by supraspinatus (suprascapular nerve).
- From 15°–90°, abduction is carried out mainly by the deltoid (axillary nerve).
- Hence, deltoid paralysis explains the patient's inability to abduct her arm beyond 15°.

iv) Sensory loss nerve:

- **Axillary nerve** – skin over the lateral side of the shoulder (regimental badge area)
- The **upper lateral cutaneous nerve of the arm** (a branch of axillary nerve) supplies skin over the **lateral side of the shoulder** – known as the **"regimental badge area."**
- Loss of sensation in this specific area is a classical feature of axillary nerve injury.







4. What is the deformity seen in Klumpke's Paralysis? Explain with its anatomical basis.

Definition:

Paralysis due to injury of the **lower trunk of brachial plexus** (C8, T1).

Nerve Involve

Klumpke's paralysis = paralysis due to injury of the **lower trunk of the brachial plexus (C8, T1 roots)**.

Lower trunk gives fibers mainly to **ulnar nerve** and partly to **median nerve**, which supply intrinsic hand muscles and medial forearm flexors.

Causes:

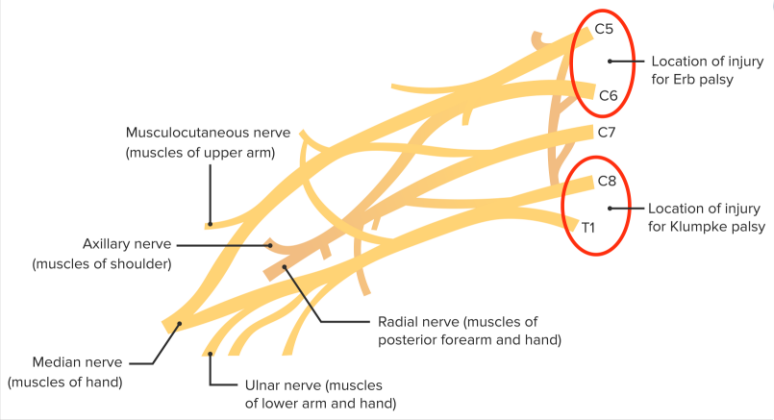
- Birth injury (excessive upward pull on arm)
- Hanging from a tree
- Catching a heavy object with a jerk

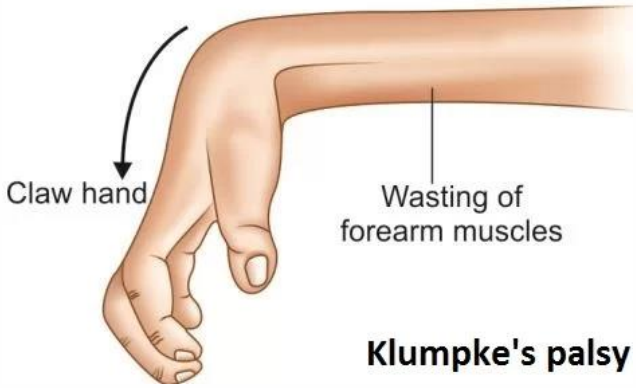
Affected Muscles:

- **Intrinsic muscles of hand**
- **Ulnar side forearm flexors**

Deformity:

- **Claw hand** – hyperextension at MCP and flexion at IP joints
- Weak grip
- Sensory loss along medial forearm and hand





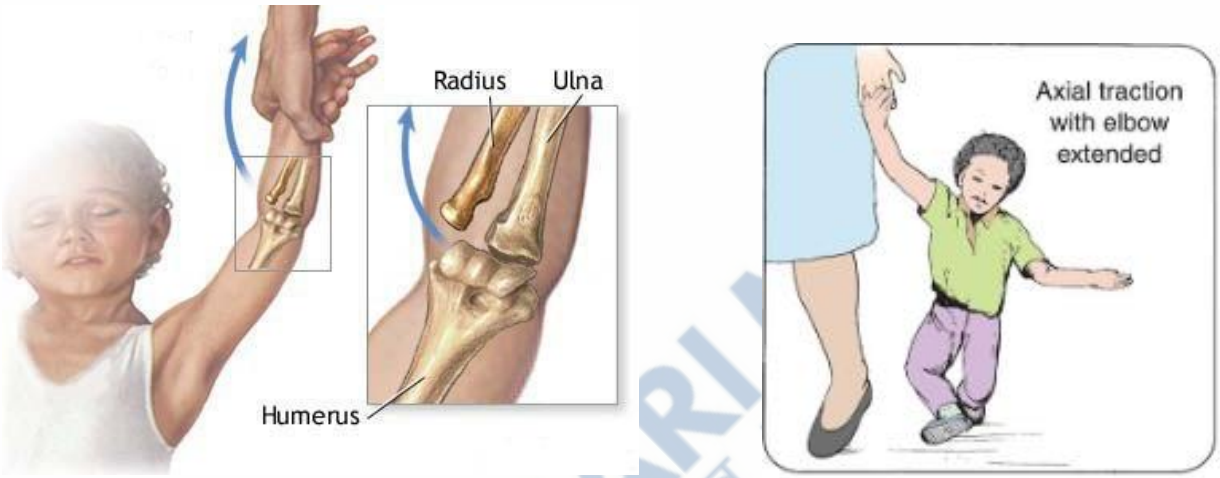
5. Subluxation of Superior Radio-Ulnar Joint in Children,Why common in children:



- In children, the **head of the radius is smaller** and the **annular ligament is more elastic**, making it easy to slip out.
- Happens when child's arm is pulled suddenly (e.g., lifting by hand)
- Also called **"nursemaid's elbow"**

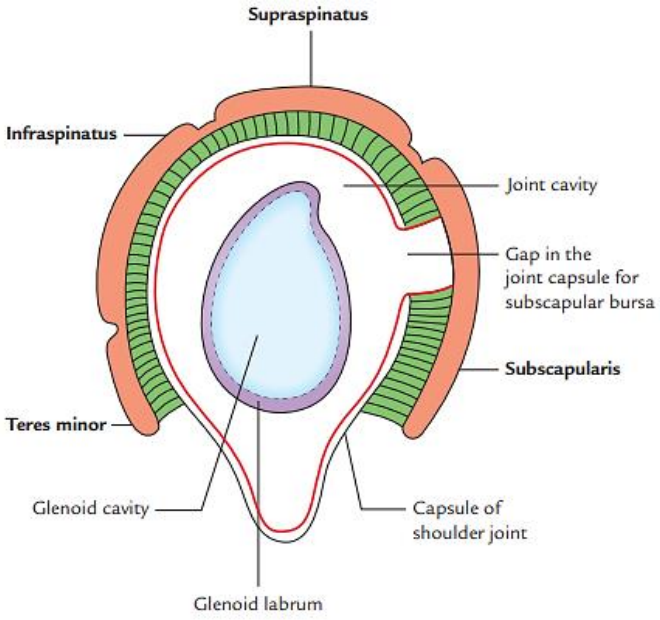
Ligaments of the joint:

1. **Annular ligament** – surrounds head of radius
2. **Quadrate ligament** – between ulna and radius neck
3. **Interosseous membrane** – connects radius and ulna shaft



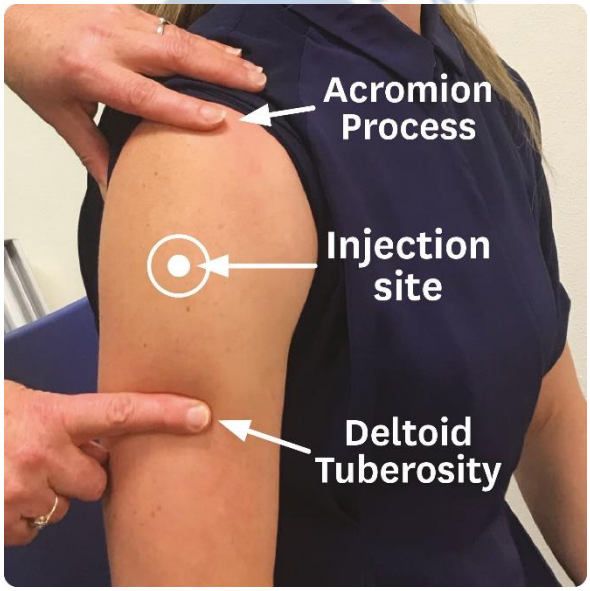
Q.6. Shoulder joint is having more mobility but less stability. Explain.

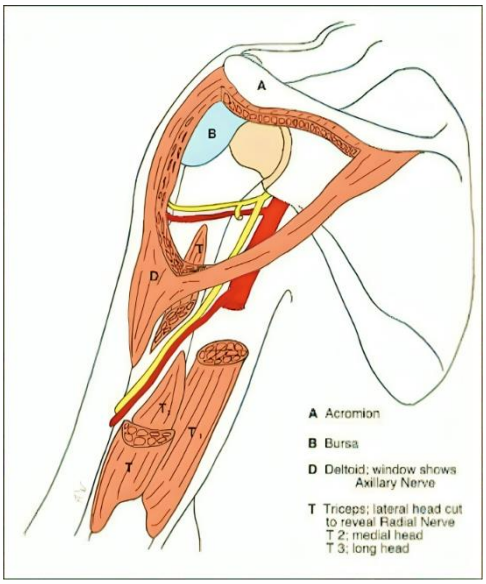
- The **shoulder joint** (glenohumeral joint) is the **most mobile joint** in the body.
- Formed between **large head of humerus** and **shallow glenoid cavity** of scapula.
- **Reasons for more mobility:**
  - Shallow socket (glenoid fossa)
  - Loose joint capsule
  - Allows wide range of motion in all directions
- **Reasons for less stability:**
  - Shallow glenoid cavity cannot hold humeral head tightly
  - Joint depends on soft tissue structures for stability:
    - **Rotator cuff muscles (SITS)**
    - **Glenoid labrum**
    - **Capsule and ligaments**
- Hence, more prone to **dislocation**.



Q.7. IM injection given in deltoid almost in middle of muscle. What complication occurs if given at other site?

- **Ideal site:**
  - Middle 1/3 of deltoid, around 2.5 to 5 cm below the **acromion** process.
- **Complications if given at wrong site:**
  - **Too high:** May injure **axillary nerve** → causes deltoid paralysis and loss of shoulder sensation.
  - **Too low or deep:** May injure **radial nerve** → leads to **wrist drop**.
  - May also injure **posterior circumflex humeral vessels**, causing bleeding.
- Proper site selection prevents **nerve damage and vascular injury**.





REASONING - UPPER LIMB

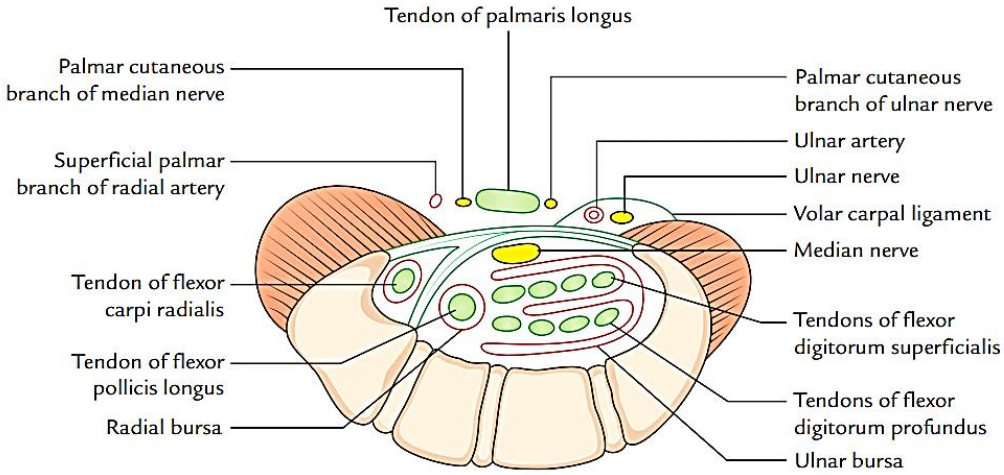
Q.8 Name the structures forming carpal tunnel. Give anatomical reasons for various symptoms in carpal tunnel syndrome.

Structures forming carpal tunnel:

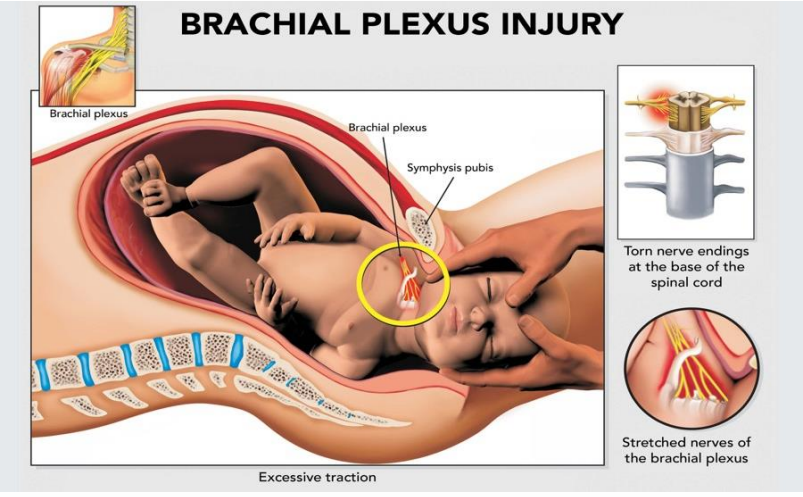
- **Roof:** Flexor retinaculum
- **Floor and sides:** Carpal bones (concave)
- **Contents:**
  - **Median nerve**
  - **Tendons of Flexor digitorum superficialis (4)**
  - **Tendons of Flexor digitorum profundus (4)**
  - **Tendon of Flexor pollicis longus**

Symptoms of carpal tunnel syndrome (anatomical basis):

- **Compression of median nerve** causes:
  - Numbness, tingling in **lateral 3½ fingers**
  - Weakness of **thenar muscles** → difficulty in thumb opposition
  - Wasting of **thenar eminence** in chronic cases
- Pain may worsen at night due to **increased pressure in tunnel**.

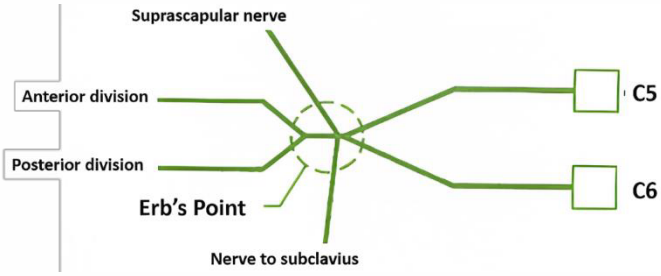


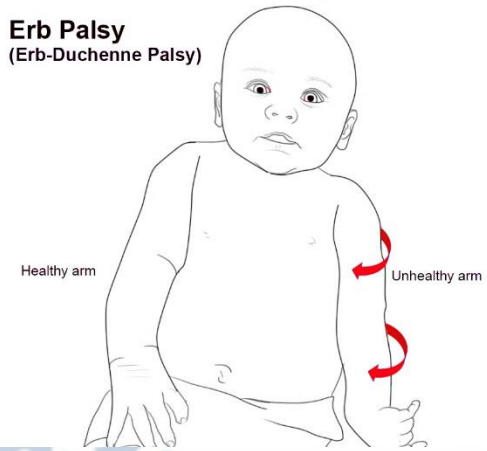
Q.9. What is the site of injury in Erb's paralysis? Name the nerves which are affected. Explain the deformity and position of the limb.



SITE OF INJURY

ERB'S POINT IN UPPER TRUNK , SIX NERVES MEET HERE





CAUSE OF INJURY

BIRTH INJURY, FALL ON SHOULDER, DURING ANAESTHESIA

NERVE ROOTS INVOLVE

MAINLY C5 AND PARTLY C6

MUSCLES PARALYSED

MAINLY BICEP BRACHII, DELTOID, BRACHIALIS AND BRACHIORADIALIS

PARTLY SUPRASPINATUS, INFRASPINATUS AND SUPINATOR

DEFORMITY - ARM - ADDUCTED AND MEDIALLY ROTATED

FOREARM - EXTENDED & PRONATED

THE DEFORMITY IS ALSO KNOWN AS POLICEMAN'S TIP HAND

DISABILITY

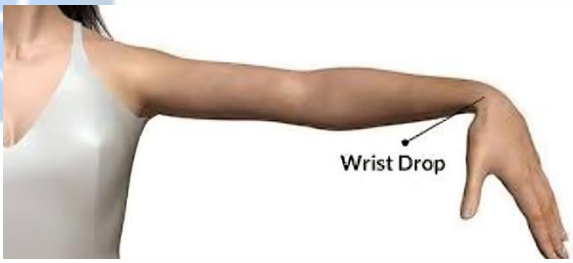
MOVEMENTS LOST - ABDUCTION & LATERAL ROTATION OF ARM & SHOULDER

- FLEXION & SUPINATION OF FOREARM

- BICEPS AND SUPINATOR JERK

Q.10 Give the anatomical basis of wrist drop due to radial nerve injury.

- **Wrist drop:**
  - Inability to extend wrist and fingers — the hand "drops" due to gravity.



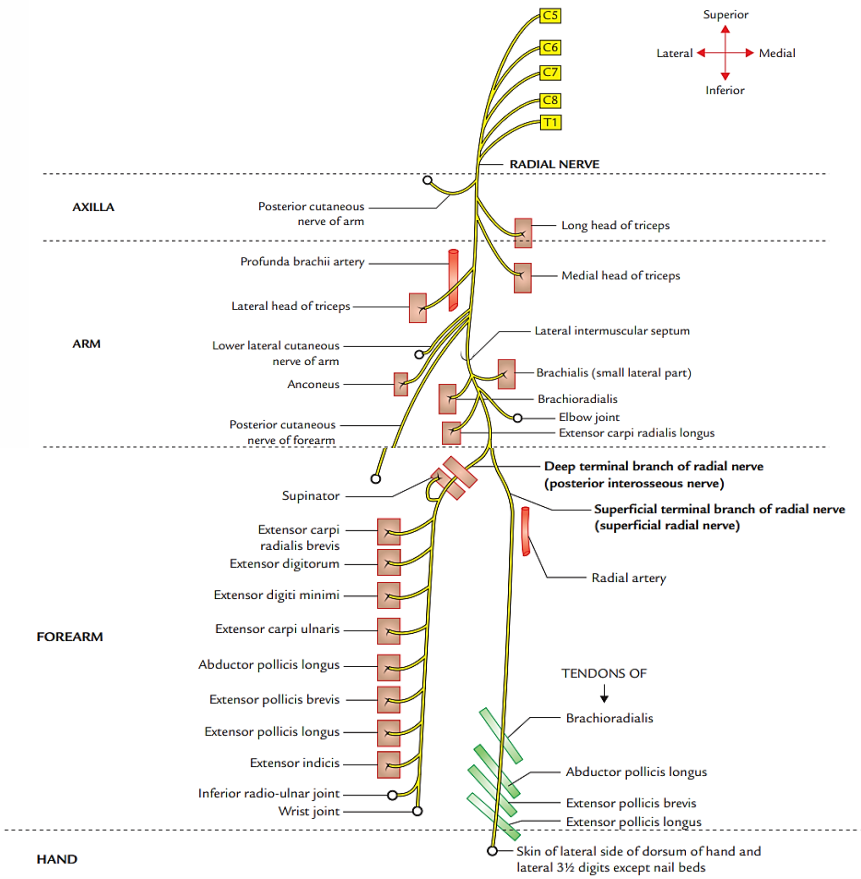
- **Cause:**
  - **Radial nerve injury**, commonly at mid-shaft of humerus (radial groove fracture)
- **Muscles affected:**
  - **Wrist extensors:** Extensor carpi radialis longus & brevis, extensor carpi ulnaris

- **Finger extensors:** Extensor digitorum, extensor pollicis

- **Anatomical basis:**
  - These extensors are supplied by the radial nerve.
  - When nerve is injured, they are paralyzed → flexors act unopposed → wrist drop.
  - Also leads to **weak grip** because wrist cannot be stabilized.

Q.11. Will radial nerve injury in cubital fossa cause wrist drop? Why?

- **No**, radial nerve injury at the **cubital fossa** will **not cause wrist drop**.
- **REASON:**
  - At the cubital fossa, the radial nerve has already given off branches to the **extensor muscles** of the forearm in the posterior compartment.
  - Only the **superficial branch** (sensory) and **deep branch (posterior interosseous nerve)** continue distally.
  - Injury here affects mainly **finger extension** and **wrist extension to some extent**, but wrist drop is usually due to injury at or above the **spiral groove of humerus**, where the radial nerve innervates the wrist extensors.
- So, wrist drop results from **radial nerve injury at the spiral groove or above**, not at cubital fossa.



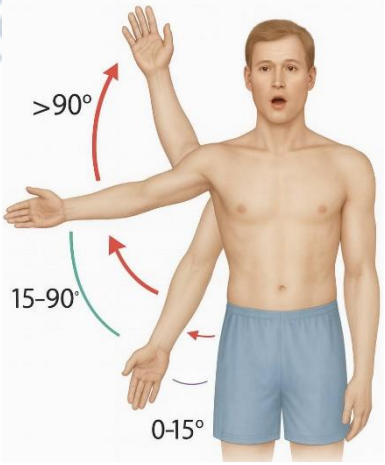
Q.12. Fracture of ulna at mid-shaft → inability to extend wrist joint

- Usually, **fracture of ulna alone** does **not cause wrist drop** because radial nerve runs in the anterior compartment and may not be injured.
- However, if the fracture is associated with injury to the **radial nerve** (which wraps around the posterior aspect of the humerus but passes anterior near the elbow), wrist extension can be lost.
- Wrist drop occurs if the **radial nerve or its branches** are injured.

Q.13 Abduction of the shoulder joint beyond 90 degrees not possible in long thoracic nerve injury. Why?

- **Long thoracic nerve** supplies the **serratus anterior muscle**.
- Serratus anterior is responsible for **rotating the scapula upward** and holding it against the chest wall.
- During abduction beyond 90 degrees, the scapula must rotate upwards for full arm elevation.
- If the long thoracic nerve is injured:
  - Serratus anterior becomes paralyzed.
  - The scapula **wings out** (scapular winging).
  - Upward rotation of scapula is lost.
  - Hence, the arm cannot be fully abducted beyond 90 degrees





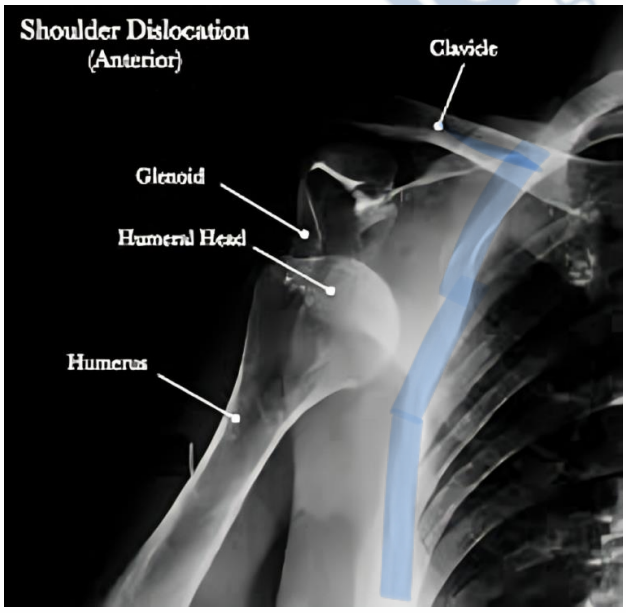
Q.14. Explain the anatomical basis of injury to axillary nerve during intramuscular injections.

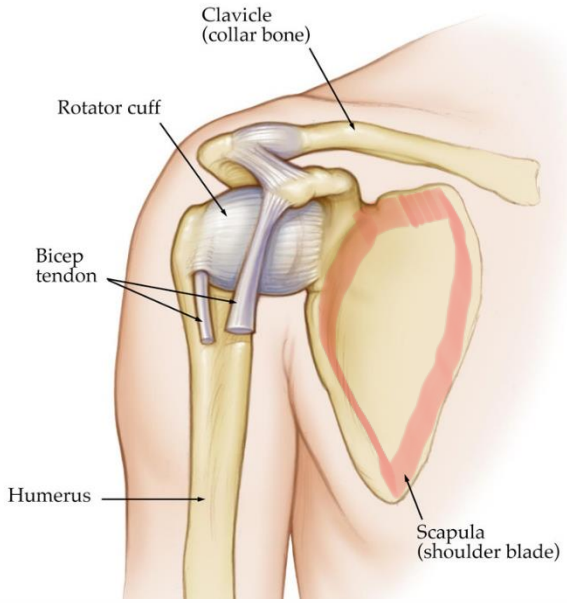
- The **axillary nerve** runs close to the **surgical neck of the humerus** and passes through the **quadrangular space**.
- The nerve is located deep to the **deltoid muscle**, approximately 5 cm below the acromion.

- Intramuscular injections given **too high** (near the acromion) or **too low** can injure the axillary nerve.
- **Improper site of injection can cause nerve damage leading to:**
  - Deltoid muscle paralysis
  - Impaired shoulder abduction beyond 15°
  - Sensory loss over the lateral shoulder
- Therefore, the recommended site for deltoid IM injection is the **middle third** of the deltoid to avoid injury.

Q.15. Why is inferior dislocation of shoulder joint common?

- The **shoulder joint** is inherently unstable due to the shallow glenoid cavity.
- **Inferior dislocation** occurs when the humeral head is forced downward out of the glenoid cavity.
- It is common because:
  - The joint capsule is weakest inferiorly.
  - The **axillary recess** (a pouch in the joint capsule) allows excessive downward movement.
  - Trauma causing forced abduction with an upward force on the humerus can push the humeral head inferiorly.
- Inferior dislocation is less common than anterior dislocation but can occur due to a fall with the arm fully abducted.
- It may cause injury to the **axillary nerve** and vessels.





Q.16. What is the anatomical basis of Ape Thumb deformity?

Definition:

Inability to oppose the thumb, thumb lies in the same plane as the fingers (ape-like hand).

Cause:

Injury to → median nerve at the wrist (e.g., carpal tunnel, trauma).

Muscles affected:

Thenar muscles – opponens pollicis, abductor pollicis brevis, flexor pollicis brevis (superficial head).

Anatomical basis:

Loss of thumb opposition and abduction → thumb pulled into the plane of palm by adductor pollicis (ulnar nerve).

Clinical feature:

- Flattening of thenar eminence
- Thumb adducted, cannot oppose



Claw hand of the first and second finger

Q.17. Explain the anatomical basis of Benediction Sign.

Definition:

When patient tries to make a fist → index & middle fingers remain extended, while ring and little finger flex.

Cause:

Median nerve injury at the elbow (supracondylar fracture).

Muscles affected:

- Flexor digitorum superficialis (all fingers)
- Lateral half of flexor digitorum profundus (index, middle)
- Thenar muscles

Anatomical basis:

Loss of flexion at MCP & IP joints of index & middle fingers, but ring/little fingers flex (ulnar nerve intact).

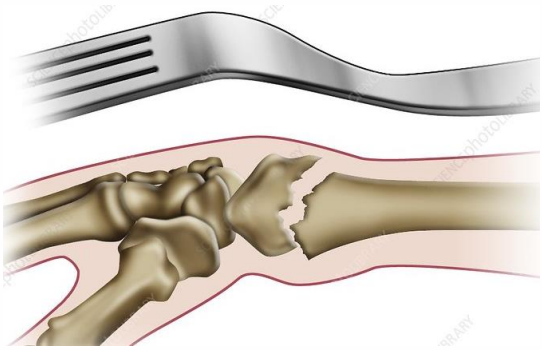
Clinical feature:

"Hand of Benediction" seen only on attempted fist.

Q.18. What is the anatomical basis of Dinner Fork Deformity in Colles' fracture?

Definition:

Backward & upward displacement of distal fragment of radius gives wrist a "dinner fork" appearance.





Cause:

Fracture of lower end of radius (2–3 cm proximal to wrist joint) after fall on outstretched hand.

Anatomical basis:

- Distal fragment displaced dorsally, laterally & proximally
- Hand deviated laterally (due to intact ulna)

Clinical feature:

- Prominent ulnar styloid
- Loss of wrist movements
- **Possible injury to median nerve (carpal tunnel compression).**

Q.19. Why is Scaphoid fracture clinically important?

Definition:

Fracture of scaphoid bone (commonest carpal fracture).

Cause:

Fall on outstretched hand.

Anatomical basis:

- Scaphoid has retrograde blood supply (distal to proximal).

Clinical feature:

- Pain in anatomical snuff box
- Delayed healing or non-union
- Avascular necrosis of proximal scaphoid.







Q.20. Explain the anatomical basis of Saturday Night Palsy.

Definition:

Temporary paralysis of radial nerve after prolonged compression in spiral groove.

Cause:

Sleeping with arm hanging over chair or under another person's head.

Muscles affected:

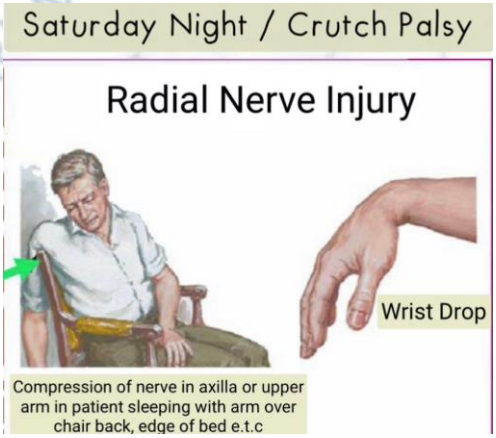
- Wrist & finger extensors paralyzed
- Triceps spared (nerve to triceps given above groove).

Anatomical basis:

Radial nerve compressed at spiral groove → loss of extension at wrist & fingers.

Clinical feature:

- Wrist drop
- Weak grip
- Sensory loss on dorsum of hand (first web space).



Q.21 What is the anatomical basis of Volkmann's Ischemic Contracture?

Definition:

Ischemic necrosis of forearm flexor muscles → fibrosis → permanent flexion deformity.

Cause:

Tight cast, compartment syndrome, brachial artery injury near elbow.

Anatomical basis:

- Flexor muscles (FDS, FDP, FPL) undergo necrosis → replaced by fibrous tissue → shortened → flexion deformity.

Clinical feature:

- Claw-like fixed flexion of fingers & wrist
- Loss of hand function
- Pain on passive extension of fingers (early sign).





Q.22. What is the anatomical basis of Drop Arm Test?

Definition:

Patient asked to slowly lower fully abducted arm → if arm drops suddenly, test is positive.

Cause:

Supraspinatus tendon tear or suprascapular nerve injury.

Anatomical basis:

Supraspinatus is essential to hold head of humerus in glenoid cavity during abduction → loss causes sudden drop.

Clinical feature:

- Arm drops at ~90° abduction
- Seen in rotator cuff injuries.

Q.23. What is the anatomical basis of Allen's Test in the upper limb?

Definition:

Allen's test checks the patency of radial & ulnar arteries at the wrist.

Procedure:

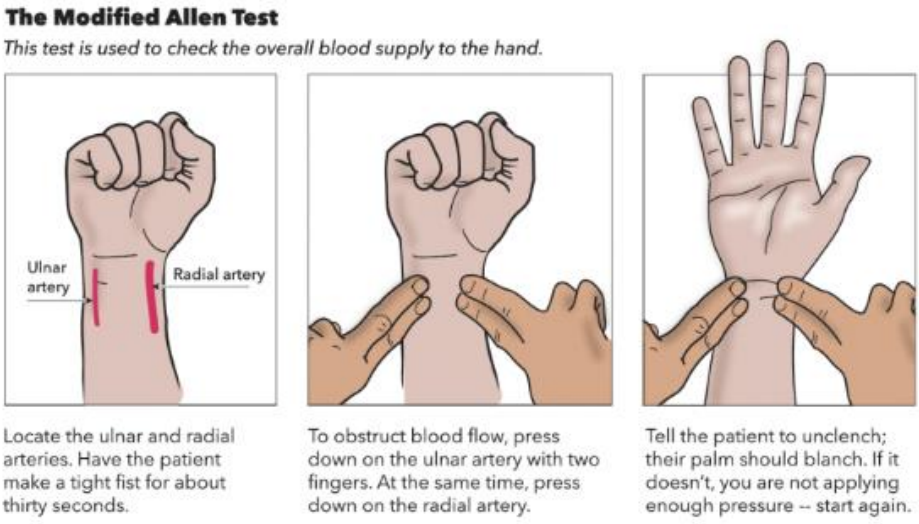
- Patient clenches fist → examiner compresses both arteries → hand turns pale.
- Release one artery → color should return within seconds.

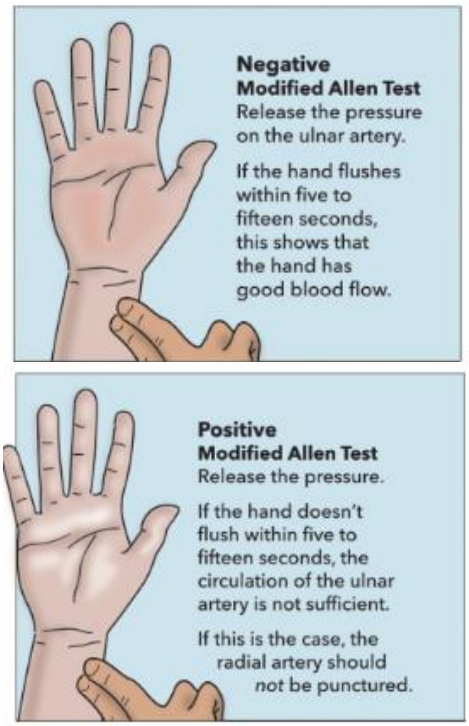
Anatomical basis:

- Hand has dual blood supply via superficial & deep palmar arches.
- If one artery is occluded and the other is insufficient → color return delayed.

Clinical importance:

- Performed before harvesting radial artery graft for CABG (Coronary Artery Bypass Grafting)





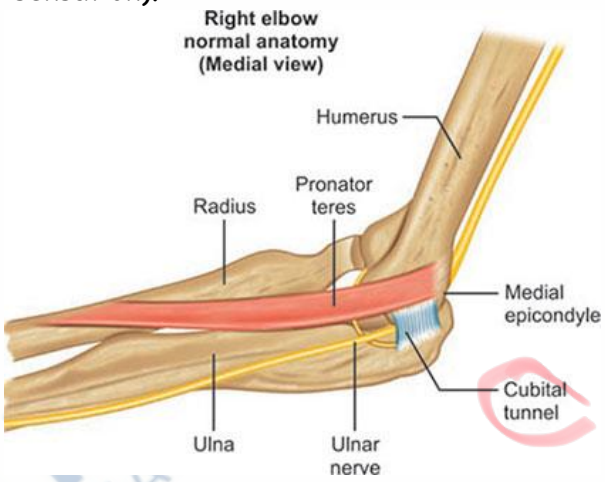
Q.24. Why is the medial epicondyle of humerus prone to ulnar nerve injury?

REASONING:

- Ulnar nerve passes **behind medial epicondyle** in a groove → superficial & unprotected.
- A blow/fracture → nerve compressed ("funny bone" sensation).

CLINICAL BASIS:

- Tingling in medial 1½ fingers
- Clawing of ring & little finger
- Weak grip strength.



Q.25. What is the anatomical basis of Waiter's Tip deformity in Erb's palsy?

DEFINITION:

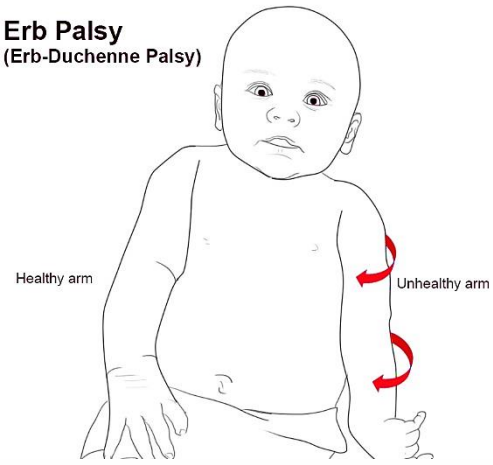
Upper brachial plexus injury (C5–C6 roots).

ANATOMICAL BASIS:

- Paralysis of: deltoid, supraspinatus, infraspinatus, biceps, brachialis.
- Arm → adducted, medially rotated
- Forearm → extended, pronated.

Why "Waiter's Tip"?

Hand hangs at side as if asking for a tip.



Q.26. Explain the anatomical basis of Parsonage-Turner Syndrome (Brachial Neuritis).

Definition:

Acute inflammation of brachial plexus → sudden severe pain → muscle weakness.

Cause:

Viral infection, post-vaccination, autoimmune.



Anatomical basis:

- Involves upper plexus (C5–C6) or sometimes entire plexus.
- Leads to weakness of shoulder & arm muscles.

Clinical features:

- Sudden severe shoulder pain
- Weakness of deltoid, supraspinatus, serratus anterior
- Wasting after weeks.

Q.27. What is the anatomical basis of Popeye Deformity?

Definition:

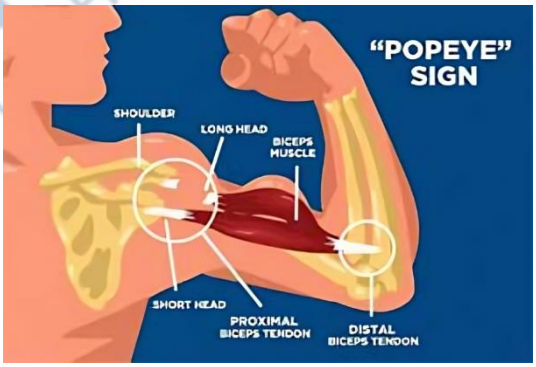
Bulging of biceps muscle belly.

Reasoning:

- Due to rupture of long head of biceps tendon at the supraglenoid tubercle.
- Muscle retracts downward → ball-like bulge in arm.
  Clinical Feature: Weak supination & flexion at elbow.







Q.28. What is theanatomical basis of Tennis Elbow (Lateral Epicondylitis)?

Definition:

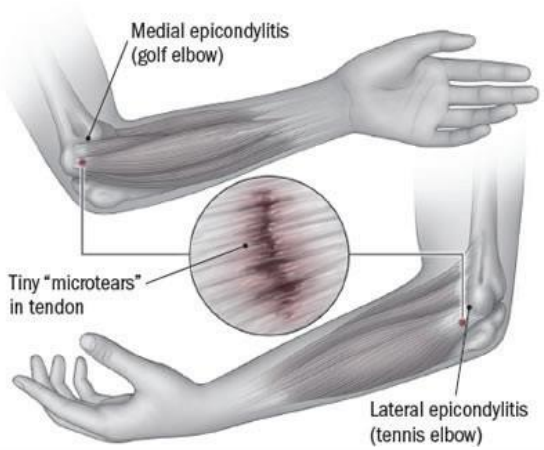
Painful condition around the lateral side of elbow.

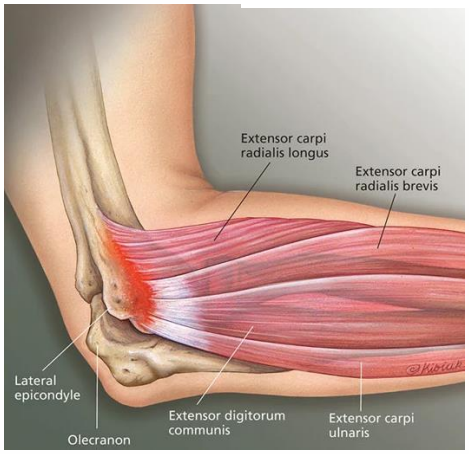
Reasoning:

- Overuse injury of extensor muscles of forearm (mainly extensor carpi radialis brevis) → microtears at their origin from the lateral epicondyle.
- Repeated pronation-supination and extension (as in tennis backhand stroke) worsens it.

Clinical Feature:

- Pain & tenderness over lateral epicondyle.
- Difficulty in gripping, wrist extension, and lifting objects.





Q.29. Why can carcinoma of breast (mammary gland cancer) present with upper limb edema and restricted movements?

Reasoning:

- Breast cancer can infiltrate and block axillary lymph nodes.
- Lymphatic obstruction → lymphedema of upper limb.
- Tumor may invade pectoral muscles or axillary wall → restriction of shoulder movements.

Clinical Feature:

- Swelling of arm, peau d'orange skin changes, limited abduction.

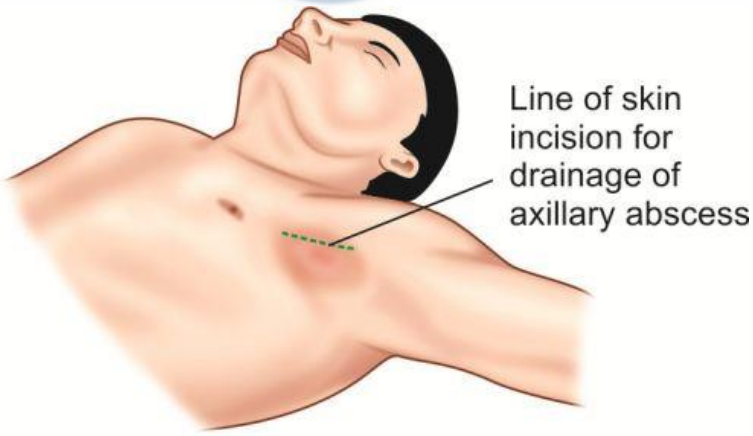
Q.30. Why is the axilla an important site for surgical drainage of abscesses?

Reasoning:

- Axilla contains axillary lymph nodes, vessels & brachial plexus.
- Abscess here may spread quickly due to loose connective tissue.
- Surgical drainage must be done carefully to avoid long thoracic nerve (serratus anterior → winging scapula) and thoracodorsal nerve (latissimus dorsi weakness).

Clinical Features:

- Swelling, fever, painful restricted shoulder movements.





ANATOMY IN 30 DAYS

BIOCHEMISTRY IN 7 DAYS



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This is a sample of notes. The new edition includes additional diagrams, questions, clinical case studies, and reasoning questions